NATURAL EXPONENTIAL AND LOGARITHMIC FUNCTIONS

DIFFERENTIAL CALCULUS

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

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60%

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SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

Derivative of Natural Exponential Function

Derivative of Natural Logarithmic Function

DERIVATIVE OF NATURAL EXPONENTIAL FUNCTION

EXPONENTIAL FUNCTION

Exponential function is a function of the form

$$f(x) = a^x$$

Laws of Exponents

$$a^{x+y} = a^x a^y$$

$$a^{x-y} = \frac{a^x}{a^y}$$

$$(a^x)^y = a^{xy}$$

$$(ab)^x = a^x b^x$$

DERIVATIVE OF e^u

$$\frac{d}{dx}(e^x) = e^x$$

Chain Rule

$$\frac{d}{dx}(e^u) = e^u \frac{du}{dx}$$

Differentiate the function $y = e^{\tan x}$

Let $u = \tan x$ $\qquad f(u) = e^u$

$$\frac{dy}{dx} = \frac{df(u)}{du} \cdot \frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{d}{du} e^u \cdot \frac{d}{dx} \tan x$$

$$\frac{dy}{dx} = e^u \cdot \sec^2 x$$

$$\boxed{\frac{dy}{dx} = e^{\tan x} \sec^2 x}$$

ans

Find y' if $y = e^{-4x}\sin 5x$.

$$\frac{dy}{dx} = \sin 5x \frac{d}{dx} e^{-4x} + e^{-4x}\frac{d}{dx}\sin 5x$$

$$\frac{dy}{dx} = \sin 5x\left(-4e^{-4x}\right) +$$

$$F(x) = e^{-4x}$$

Let $u = -4x$ $\quad f(u) = e^{u}$

$$F'(x) = \frac{df(u)}{du}\cdot\frac{du}{dx}$$

$$F'(x) = \frac{d}{du}e^{u}\cdot\frac{d}{dx}(-4x)$$

$$F'(x) = e^{u}\cdot(-4)$$

$$F'(x) = -4e^{-4x}$$

Find y' if $y = e^{-4x}\sin 5x$.

$$\frac{dy}{dx} = \sin 5x \frac{d}{dx} e^{-4x} + e^{-4x} \frac{d}{dx} \sin 5x$$

$$\frac{dy}{dx} = \sin 5x \left(-4e^{-4x}\right) +$$

$$F(x) = \sin 5x$$

Let $u = 5x$ $\quad f(u) = \sin u$

$$\frac{dy}{dx} = \frac{df(u)}{du} \cdot \frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{d}{du} \sin u \cdot \frac{d}{dx}(5x)$$

$$\frac{dy}{dx} = \cos u \cdot (5)$$

$$\frac{dy}{dx} = 5\cos 5x$$

Find y' if $y = e^{-4x}\sin 5x$.

$$\frac{dy}{dx} = \sin 5x \frac{d}{dx} e^{-4x} + e^{-4x} \frac{d}{dx} \sin 5x$$

$$\frac{dy}{dx} = \sin 5x \left(-4e^{-4x}\right) + e^{-4x}(5\cos 5x)$$

$$\boxed{\frac{dy}{dx} = e^{-4x}(5\cos 5x - 4\sin 5x)}$$

ans

$F(x) = \sin 5x$

Let $u = 5x$ $\quad f(u) = \sin u$

$$\frac{dy}{dx} = \frac{df(u)}{du} \cdot \frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{d}{du}\sin u \cdot \frac{d}{dx}(5x)$$

$$\frac{dy}{dx} = \cos u \cdot (5)$$

$$\frac{dy}{dx} = 5\cos 5x$$

DERIVATIVE OF NATURAL LOGARITHMIC FUNCTIONS

LOGARITHMIC FUNCTION

Logarithmic function is a function of the form

$$\log_a x = y$$

Natural logarithm is a logarithm with **base e**

$$\log_e x = \ln x$$

Laws of Logarithms

$$\log_a(xy) = \log_a x + \log_a y$$

$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$

$$\log_a(x^r) = r\log_a x$$

DERIVATIVE OF $\ln u$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

Chain Rule

$$\frac{d}{dx}(\ln u) = \frac{1}{u}\frac{du}{dx}$$

Differentiate the function $y = \ln(\sin x)$

Let $u = \sin x$ $\qquad f(u) = \ln u$

$$\frac{df(u)}{du} = \frac{1}{u} \qquad \frac{du}{dx} = \cos x$$

$$\frac{dy}{dx} = \frac{1}{u} \cdot \cos x$$

$$\frac{dy}{dx} = \frac{1}{\sin x} \cdot \cos x$$

$$\boxed{\frac{dy}{dx} = \cot x}$$

ans

Differentiate $f(x) = \sqrt{\ln x}$.

Let $u = \ln x$ $\qquad f(u) = \sqrt{u}$

$$\frac{df(x)}{dx} = \frac{df(u)}{du} \cdot \frac{du}{dx}$$

$$\frac{df(x)}{dx} = \frac{d}{du}\sqrt{u} \cdot \frac{d}{dx}\ln x$$

$$\frac{df(x)}{dx} = \frac{1}{2}u^{-\frac{1}{2}} \cdot \frac{1}{x}$$

$$\frac{df(x)}{dx} = \frac{1}{2x\sqrt{u}}$$

$$\boxed{\frac{df(x)}{dx} = \frac{1}{2x\sqrt{\ln x}}}$$

ans

SEATWORK